

On the Synthesis of Optimum Linear Feedback Control Systems

by W. C. PETERSON and A. H. NASSAR

Department of Electrical Engineering, North Carolina State University, Raleigh, Raleigh, NC 27650, U.S.A.

ABSTRACT: Standard forms are presented which define transfer functions for optimum type 1 and type 2 feedback control systems, where systems with minimum integral of time multiplied absolute value of error in the presence of a step displacement input are considered to be optimum. The ITAE criterion adopted in this paper was previously introduced by Graham and Lathrop.

The optimization procedure leading to the standard forms presented here, is based upon an all digital simulation coupled to an unconstrained optimization algorithm to minimize the ITAE criterion value.

It is shown with the aid of examples how actual systems can be compensated by use of appropriate standard forms to obtain optimum responses.

Nomenclature

$A_1 \dots A_{n-1}$	system coefficient values
$c(t)$	system output
$C(s)$	transform of system output
$E_i(s)$	input voltage transform
$E_o(s)$	output voltage transform
$F(s)$	system overall function
$G(s)$	system loop function
$G_c(s)$	compensator function
$G_p(s)$	plant function
$H(s)$	feedback function
n	system order
P	negative of system pole, per sec
$r(t)$	system input
$R(s)$	transform of system input
s	Laplace variable
z	negative of system zero
α	system parameter
β	system parameter
$\epsilon(t)$	system error
ω_0	natural radian frequency

I. Introduction

The design of a feedback control system generally starts with a plant, that is, a process or machine which is to be controlled. A plant function ordinarily

expresses the plant characteristics. Feedback is introduced around the plant. In order to obtain acceptable system performance, it is usually found necessary to add compensation, either in cascade with the plant, or in a minor feedback path. In conventional design and compensation procedures, trial and error are often used with analytical techniques involving root locus plots or Bode or Nyquist diagrams to obtain a design which meets a given set of specifications.

The idea that once a system configuration is chosen, we can identify the system parameters and then choose values for these parameters which result in an optimum response, provides the designer with a different outlook. Under these conditions, we are evidently not satisfied to obtain a system which simply meets a set of specifications, but we endeavor to obtain a design which produces the best possible results with a given configuration, or in other words an optimum design. In this paper we consider methods for designing or synthesizing optimum feedback systems.

II. Error Function Criteria

Important papers were written by Graham and Lathrop (1-3), discussing the general ideas of system response and the various criteria which could be used to evaluate response. Much attention in Ref. (1) was directed toward minimizing some function of error where error is defined as the desired value of output minus the actual output. Some error functions considered were integral of squared error (ISE), integral of absolute error (IAE), integral of time multiplied by squared error (ITSE), integral of squared time multiplied by squared error (ISTSE), integral of squared time multiplied by absolute value of error (ISTAЕ) and integral of time-multiplied absolute value of error (ITAE). On the basis of the demonstrated superiority of the ITAE criterion when compared to all others from the standpoints of ease of applicability, clear selectivity of optima and reliability for the class of systems considered, Graham and Lathrop (1) have succeeded in presenting strong arguments for the use of the ITAE criterion as the most suitable measure of system performance. We also consider the ITAE criterion to be the most suitable, and utilize it in this paper, where the dynamic process under control is linear and the unit step function of input displacement is the test signal considered for follower type systems. The design of optimum regulator systems is also briefly considered.

The ITAE criterion is expressed by the integral

$$J = \int_0^{\infty} t |\varepsilon(t)| dt \quad (1)$$

where $\varepsilon(t)$ represents error. It may be noted that the t representing time under the integral sign has the effect of penalizing long duration errors. This is the time weighting feature.

The error $\varepsilon(t)$ in general is the desired value of the output minus the actual output, but since the input of a follower type system is generally the desired

output, we can write $\varepsilon(t) = r(t) - c(t)$ where $r(t)$ and $c(t)$ are input and output respectively.

The system for which the J value (also called the ITAE value) of Eq. (1) is a minimum for a step displacement input, is then considered to be the optimum system.

III. Basis for Standard Forms

It will be considered that the overall function for the systems of interest here may be expressed in the form (1)

$$\frac{C(s)}{R(s)} = \frac{\alpha A_{n-1} \omega_0^{n-1} s + \omega_0^n}{s^n + A_1 \omega_0 s^{n-1} + \dots + A_{n-1} \omega_0^{n-1} s + \omega_0^n} \quad (2)$$

where $A_1 \dots A_{n-1}$, α and ω_0 are system parameters, and n is the system order. In particular ω_0 is a frequency parameter called the natural radian frequency. In general, for any assumed α , values of $A_1 \dots A_{n-1}$ can be chosen so as to produce a minimum J value of Eq. (1) when the input $r(t) = L^{-1}[R(s)]$ is a step function. Note that $C(s)$ is the transform of system output, and on the basis that the input is also the desired value of the output, the transform of error is

$$\varepsilon(s) = R(s) - C(s).$$

It should be noted that no finite value of ω_0 produces a minimum J and in fact J continually decreases as ω_0 increases. For this reason, the value of ω_0 is not chosen until near the end of a design or optimization procedure. To obtain the smallest possible J , the largest possible ω_0 is used, but some physical constraint will generally establish some upper limit for ω_0 .

Since ω_0 does not enter into the optimization procedure, it was decided for convenience and ease of problem manipulation that in the process of optimization, all systems were considered to have a unity ω_0 value. We lose no generality as a result of this.

We have taken the unit step function as a convenient test input. However the application of other test inputs such as the ramp function could also produce useful results.

On the basis that $\omega_0 = 1$, we obtain the normalized transfer function

$$\frac{C(s)}{R(s)} = \frac{\alpha A_{n-1} s + 1}{s^n + A_1 s^{n-1} + \dots + A_{n-1} s + 1}. \quad (3)$$

If we write the last function in the form

$$\frac{C(s)}{R(s)} = \frac{\alpha A_{n-1} s + 1}{s^n + A_1 s^{n-1} + \dots + A_{n-1} (1 - \alpha) s + \alpha A_{n-1} s + 1} \quad (4)$$

and then set

$$\frac{C(s)}{R(s)} = F(s)$$

we see that for a unity feedback system with a forward function $G(s)$,

$$F(s) = \frac{G(s)}{1 + G(s)}$$

from which

$$G(s) = \frac{F(s)}{1 - F(s)}. \quad (5)$$

If we now determine $G(s)$ for the $F(s) = C(s)/R(s)$ of Eq. (4), we obtain

$$G(s) = \frac{\alpha A_{n-1}s + 1}{s[s^{n-1} + A_1s^{n-2} + \dots + A_{n-1}(1 - \alpha)]}. \quad (6)$$

It is well known that the system type equals the number of integrations in the $G(s)$ function. Thus we see that for the α range $0 < \alpha < 1$, the system is type 1, but that when $\alpha = 1$, the system becomes type 2.

The denominator polynomial of Eq. (2) is the standard form for a particular value of α when $A_1, A_2, \dots, A_{n-1}$ are chosen to make J a minimum. The values of the denominator coefficients $A_1, A_2, \dots, A_{n-1}$ are of course those for an optimum system. For α values between zero and unity, the denominator polynomial with its optimum coefficients can be specified for each designated α value, or alternatively α can be plotted vs the optimum coefficients.

The Optimization Procedure

Graham and Lathrop (1) presented standard forms for systems with $\alpha = 0$ and $\alpha = 1$. We considered it desirable to repeat some of the Graham and Lathrop optimization runs thinking that this would lead only to a verification of the Graham and Lathrop values. This turned out to be the case for the lower ordered systems, but it was found that for higher order systems and all type 2 systems, different standard forms were obtained.

After normalizing each system transfer function, CSMP III (5) structure statements were used for simulating the system under test. The modeled system together with the Powell algorithm (6) thus constituted the programming body for minimizing the J value of Eq. (1). The Butterworth, binomial and some Graham and Lathrop (1, 2) standard form coefficients served in most cases as relatively good initial values to start the iteration. A 15 sec integration time was used in evaluating the J value of Eq. (1) for type 1 systems.

At the end of a computation, a set of parameters corresponding to the minimum ITAE criterion value results, which defines the optimum system coefficients. Transfer function denominators with the optimum coefficients thus obtained, are tabulated and referred to as standard forms.

Other algorithms (7, 8) were carefully examined along with Powell's, but the latter provided an appealing compromise between high convergence rates and low computing times, therefore it was the one selected.

The fact that our optimization runs produced some new standard forms must

be considered reasonable in view of the improvements in optimization procedures since the time Graham and Lathrop did their work. In fact, it seems remarkable that Graham and Lathrop obtained results as good as they did by use of only the analog computer in their study.

We expect to present details of our optimization procedure in a different paper.

IV. Some Standard Forms

The $C(s)/R(s)$ function for type 1 systems with $\alpha = 0$ can be written in the form

$$\frac{C(s)}{R(s)} = \frac{\omega_0^n}{s^n + A_1\omega_0s^{n-1} + \dots + A_{n-1}\omega_0^{n-1}s + \omega_0^n}. \quad (7)$$

The optimum system denominators, which are then the standard forms for $2 \leq n \leq 7$, become those in Table I.

TABLE I
Standard forms for $\alpha = 0$ (type 1 systems)

$s^2 + 1.4\omega_0s + \omega_0^2$
$s^3 + 1.73\omega_0s^2 + 2.15\omega_0^2s + \omega_0^3$
$s^4 + 1.87\omega_0s^3 + 3.26\omega_0^2s^2 + 2.59\omega_0^3s + \omega_0^4$
$s^5 + 1.95\omega_0s^4 + 4.4\omega_0^2s^3 + 4.53\omega_0^3s^2 + 3.22\omega_0^4s + \omega_0^5$
$s^6 + 1.15\omega_0s^5 + 5.22\omega_0^2s^4 - 4.83\omega_0^3s^3 + 6.21\omega_0^4s^2 + 3.35\omega_0^5s + \omega_0^6$
$s^7 + 1.54\omega_0s^6 + 6.22\omega_0^2s^5 + 7.35\omega_0^3s^4 + 10.6\omega_0^4s^3 + 7.73\omega_0^5s^2 + 4.19\omega_0^6s + \omega_0^7$

For type 2 systems with $\alpha = 1$, the $C(s)/R(s)$ function can be expressed as

$$\frac{C(s)}{R(s)} = \frac{A_{n-1}\omega_0^{n-1}s + \omega_0^n}{s^n + A_1\omega_0s^{n-1} + \dots + A_{n-1}\omega_0^{n-1}s + \omega_0^n}. \quad (8)$$

For such systems, the optimum system denominators which are then the standard forms for $2 \leq n \leq 6$ become those in Table II. For type 2 systems an integration time of 10 sec was used in the optimization procedure.

TABLE II
Standard forms for $\alpha = 1$ (type 2 systems)

$s^2 + 350\omega_0s + \omega_0^2$
$s^3 + 42\omega_0s^2 + 360.3\omega_0^2s + \omega_0^3$
$s^4 + 12.6\omega_0s^3 + 100.9\omega_0^2s^2 + 312.2\omega_0^3s + \omega_0^4$
$s^5 + 20.3\omega_0s^4 + 183.1\omega_0^2s^3 + 870.1\omega_0^3s^2 + 1718.3\omega_0^4s + \omega_0^5$
$s^6 + 5.96\omega_0s^5 + 43.97\omega_0^2s^4 + 142.66\omega_0^3s^3 + 328\omega_0^4s^2 + 325.3\omega_0^5s + \omega_0^6$

The output responses $C(t)$ when the input is a unit step are presented for $\alpha = 0$ and $\alpha = 1$ for $2 \leq n \leq 5$ in Figs. 1–4. These curves illustrate the remarkable improvement in response obtained in changing the system from type 1, $\alpha = 0$ to type 2, $\alpha = 1$. This improvement is also indicated by a reduction by factors of 10–100 in the respective ITAE values.

A pattern of similar and progressive features was observed in the sub classes of linear systems considered. This is illustrated in Fig. 5 where a consistent trend could be clearly observed in the location of optimum poles for type 1 systems with respect to system order. The higher the system order, the closer the pole locations get to the imaginary axis. For these type 1 systems, it is evident from Fig. 5 that all optimum poles are complex conjugates except that all odd ordered systems have one real pole, always the furthest from the origin.

The effect of varying ω_0 on the ITAE value is shown in Fig. 6. It is noted that the ITAE value continually decreases as ω_0 increases.

It may be noted that the upper integration limit in Eq. (1) for the J value or ITAE criterion value was infinity. For our computations it was of course necessary to replace this infinity by a finite value. Initially we used a 10 sec limit.

In a private correspondence, Dr. R. C. Lathrop suggested that the 10 sec integration limit should be increased. It was found upon reconsideration of this matter that one of our type 1 systems had a transient settling time greater than 10 sec. For this reason the integration limit for type 1 systems was changed to 15 sec. Runs were made using limits of 20 and 30 sec, and with such limits, the changes in optimum coefficient values were no more than 1%, for a type 1 system. It was concluded that the 15 sec limit was quite satisfactory for type 1 systems.

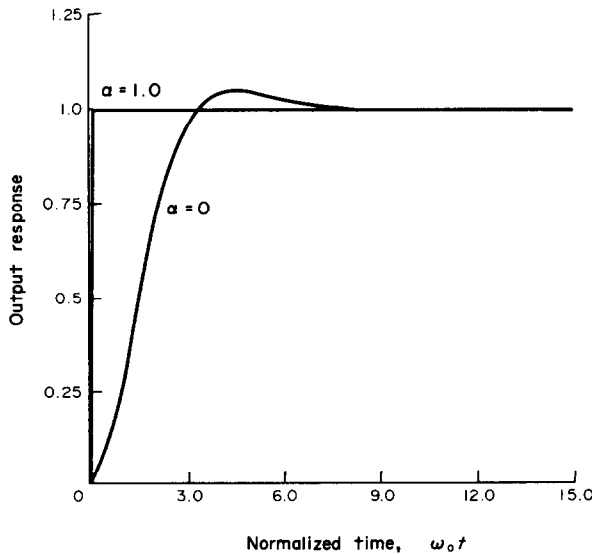


FIG. 1. Output responses $c(t)$ when the input $r(t)$ is a unit step. Second order system, types 1 and 2.

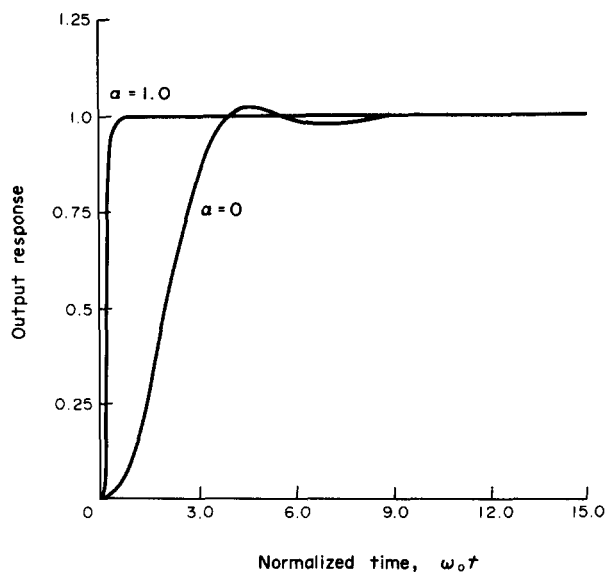


FIG. 2. Output responses $c(t)$ when the input $r(t)$ is a unit step. Third order system, types 1 and 2.

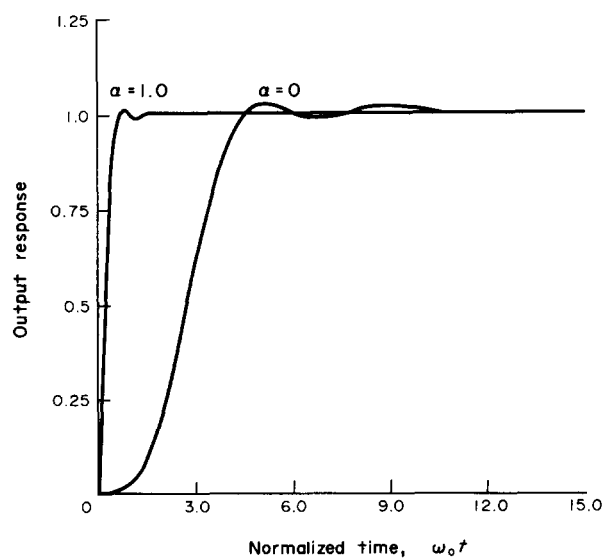


FIG. 3. Output responses $c(t)$ when the input $r(t)$ is a unit step. Fourth order system, types 1 and 2.

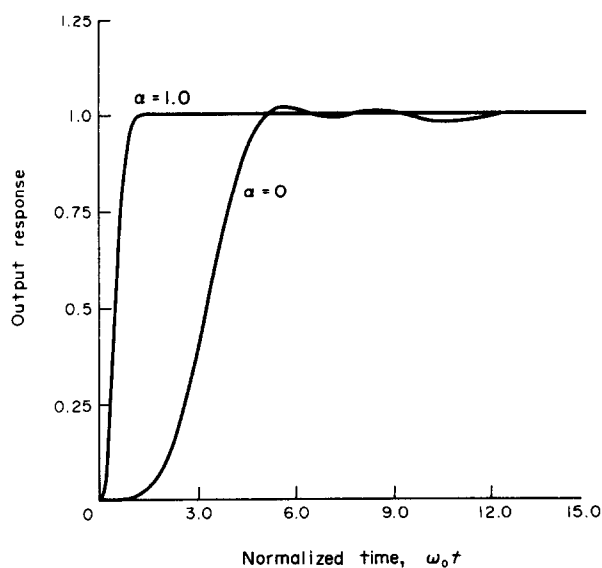


FIG. 4. Output responses $c(t)$ when the input $r(t)$ is a unit step. Fifth order system, types 1 and 2.

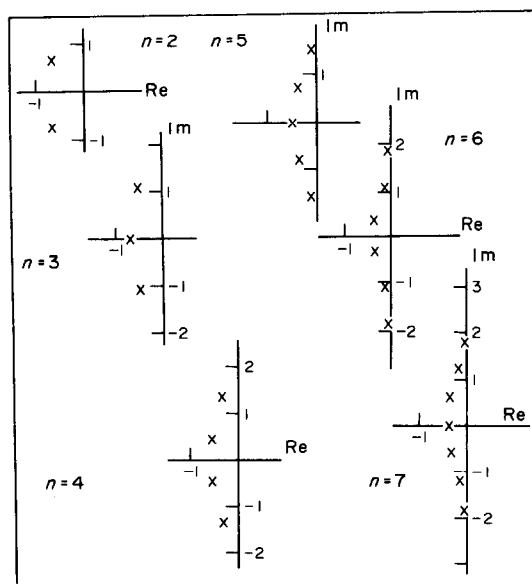


FIG. 5. Closed loop pole zero patterns for type 1 optimum systems, second to seventh order.

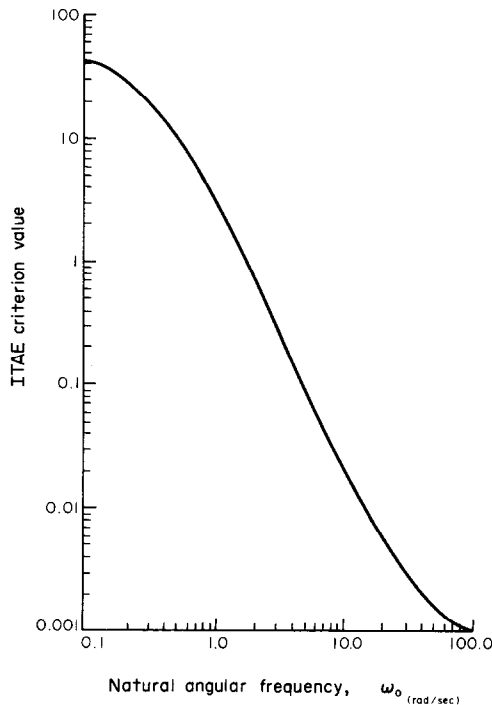


FIG. 6. Effect of varying ω_0 on the ITAE criterion value, third order type 1 optimum system.

For type 2 systems, the 10 sec integration time was in every case much larger than the transient interval. For this reason it was concluded that the 10 sec limit was quite satisfactory for type 2 systems.

V. Parameters for Intermediate α Values

It will be recalled that α values in the range $0 \leq \alpha < 1$ in Eq. (2) result in type 1 systems. Each α value produces a unique set of optimum $A_1 \dots A_{n-1}$ values. The case where $\alpha = 0$ has already been considered. For the intermediate range $0 < \alpha < 1$, sets of optimum $A_1 \dots A_{n-1}$ values corresponding to discrete α values could have been presented, but then interpolation would be needed to obtain parameter values between the discrete α values. For this reason, it was decided to present curves of α vs optimum parameter values, as shown in Figs. 7–10. The limiting values are those for $\alpha = 0$ and $\alpha = 1$, already considered, for which output responses were given in Figs. 1–4. The output responses for the range $0 < \alpha < 1$ will lie between these limiting values. In particular we note that the response of a type 1 system with an α approaching unity, will approach that of a type 2 system.

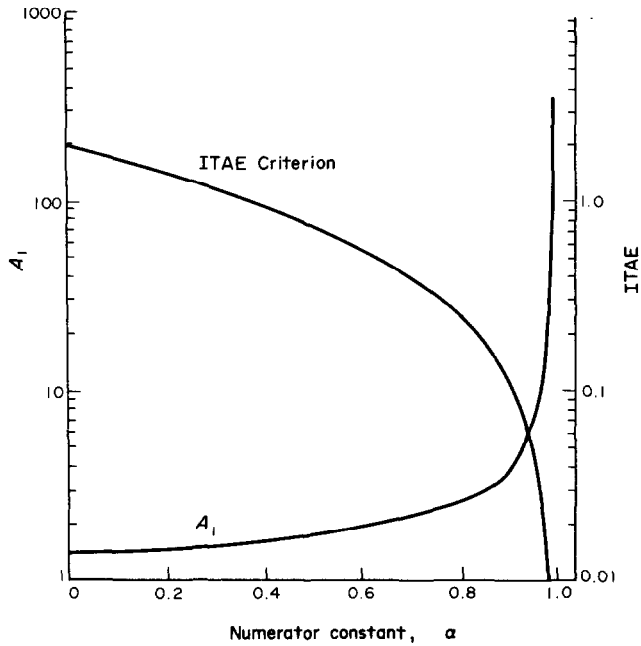


FIG. 7. Optimum parameter A_1 and ITAE criterion values for normalized second order systems, $0 \leq \alpha \leq 1$.

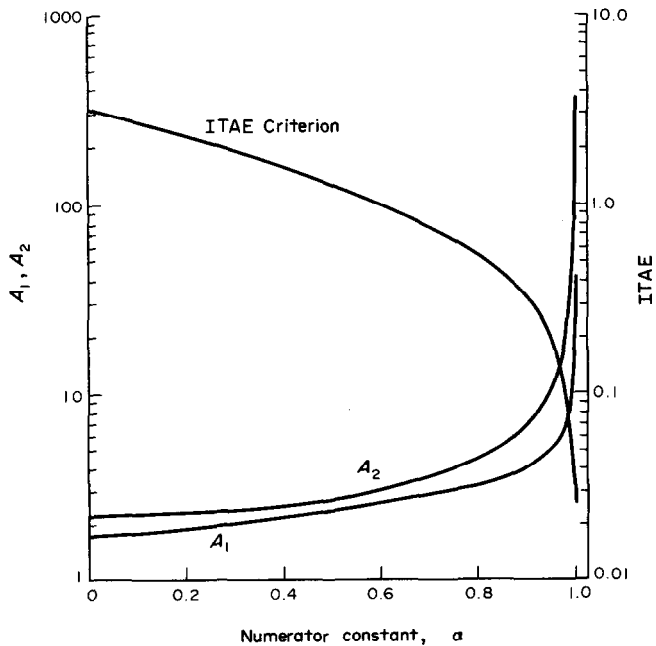


FIG. 8. Optimum parameters and ITAE criterion values for normalized third order systems, $0 \leq \alpha \leq 1$.

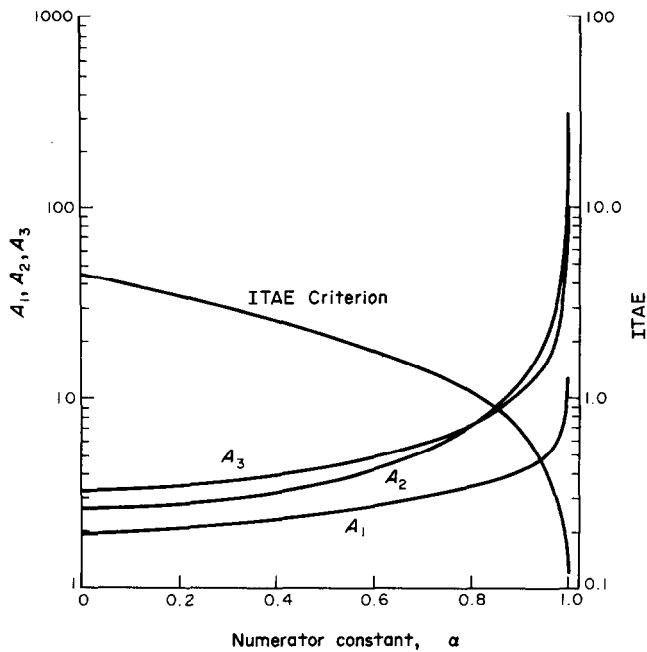


FIG. 9. Optimum parameters and ITAE criterion values for normalized fourth order systems, $0 \leq \alpha \leq 1$.

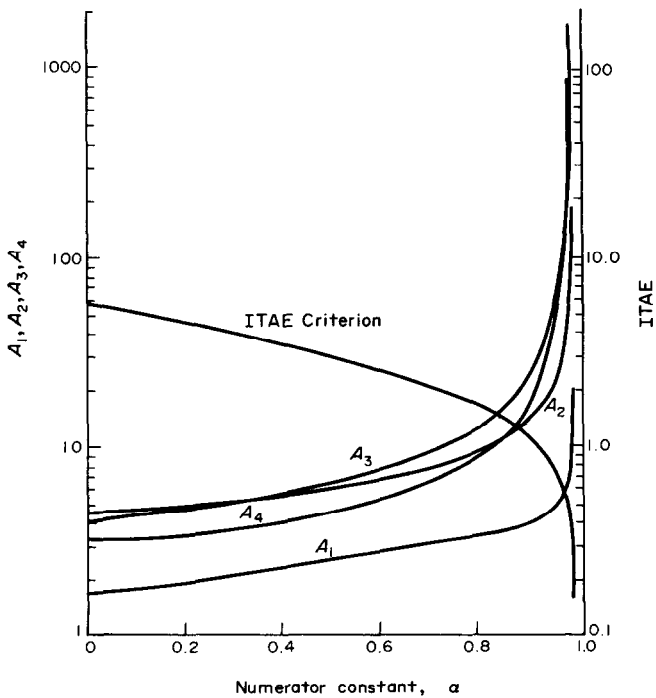


FIG. 10. Optimum parameters and ITAE criterion values for normalized fifth order systems, $0 \leq \alpha \leq 1$.

The information available from Figs. 7–10 may be considered to define standard forms for different values of α .

Note that an ITAE value is given for each value of α in Figs. 7–10. These are the ITAE values for the normalized systems with $\omega_0 = 1$. The ITAE values for the terminal α values of zero and unity are respectively the ITAE values for type 1 systems with $\alpha = 0$ and for type 2 systems. If the ITAE value is desired for a system with $\omega_0 \neq 1$, the information given in Fig. 6 can be used together with the value obtained from Figs. 7–10, to determine such a value.

VI. Some Comments on Optimum Design

The introduction of feedback around a plant function will not generally result in a system with a satisfactory response. Compensation must usually be added in cascade with the plant or in a minor feedback loop. In this paper, we propose that the compensation should be chosen so as to produce an optimum response.

If we use the ITAE criterion as a basis for optimum design, we are considering that the system with the smallest ITAE criterion value is the best one. To obtain such a system, we can determine the optimum parameters, but it should be noted that we have generally chosen a system configuration; hence what we have obtained are only the optimum parameters for a system of a given configuration. Strictly speaking, the designer should consider different configurations and finally choose the one which results in the smallest ITAE value.

Another matter to consider is the fact that the ITAE value continually decreases as ω_0 increases. For this reason, it is sometimes found that a certain system configuration which allows choosing a larger ω_0 value, will yield a smaller ITAE value than some other configuration for which the upper limit on ω_0 is smaller.

There may be situations in which it is decided that a system which is to be designed should be type 1 say, or type 2. However, if such a decision is made, we must admit that the basis of choice is no longer simply to obtain a minimum ITAE value as we suggested in the beginning. We can say that the end result in such case may be an optimum type 1 system, which is not necessarily the one with a minimum ITAE value when compared with the other systems which could be built.

It is quite evident that the ITAE value tends to decrease as system type number increases, say in going from optimum type 1 to type 2. With respect to type 1 systems, the ITAE value decreases as α is increased towards unity. Thus it seems that to make the ITAE value as small as possible, one should always make the system type 2. However, it will generally be found that the type 2 system will be more difficult to realize and build than a type 1 system. A system zero with a rather large time constant is required and generally successful design of a type 2 system requires starting with a plant for which all time constants are small. Quite often for such reasons, a decision to build a type 1 system will be made. Having made this decision, there will then be some

incentive to make α as large as possible within the range $0 < \alpha < 1$, in order to obtain a type 1 system with the smallest possible ITAE value. However, it will generally be found that the system becomes more and more difficult to realize and build as α is increased toward unity, although not as difficult as a type 2 system. Evidently some engineering judgement is required to determine how large a system type number or α value results in a practically realizable system. It must be admitted then that the basis for the final design is not entirely the ITAE criterion, although this was the original intent, but is to some extent engineering judgement.

VII. Some Design Examples

It will now be shown how the standard forms and curves can be used to design optimum feedback systems.

Example 1

Figure 11 is considered to be a fairly general configuration for a feedback control system. $G_p(s)$ is the plant function, $G_c(s)$ is cascade compensation and $H(s)$ feedback compensation. We will assume in this example that only cascade compensation is to be used, thus $H(s) = 0$. The disturbance D is also assumed to be zero.

Assume the plant function to be

$$G_p(s) = \frac{10}{s(s+1)(s+10)} \quad (9)$$

and the objective is to determine $G_c(s)$.

If we assume the system to be third order with $\alpha = 0$, then by reference to the standard forms of Table I, we have

$$F(s) = \frac{C(s)}{R(s)} = \frac{\omega_0^3}{s^3 + 1.73\omega_0 s^2 + 2.15\omega_0^2 s + \omega_0^3}.$$

Now since $F(s) = G(s)/[1 + G(s)]$, the forward function $G(s) = G_c(s)G_p(s)$ becomes

$$G(s) = \frac{F(s)}{1 - F(s)} = \frac{\omega_0^3}{s(s^2 + 1.73\omega_0 s + 2.15\omega_0^2)} \quad (10)$$

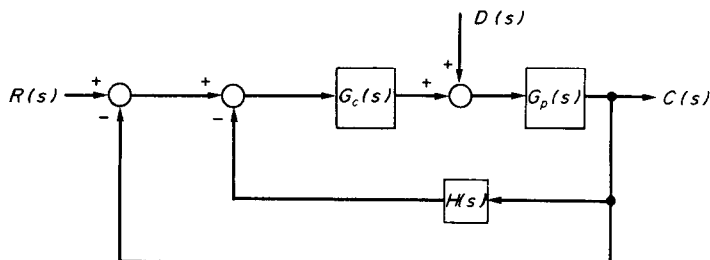


FIG. 11. Block diagram for a feedback control system.

which then results in

$$G_c(s) = \frac{G(s)}{G_p(s)} = \frac{\omega_0^3(s+1)(s+10)}{s^2 + 1.73\omega_0 s + 2.15\omega_0^2}.$$

It is noted that the denominator zeros in $G_c(s)$ are the complex values $s = -\omega_0(0.87 \pm j1.18)$, thus $G_c(s)$ cannot be realized by use of a passive network. For this reason the standard lead, lag or lead-lag networks cannot be used. However, $G_c(s)$ can be realized by use of an active network, such as an operational amplifier circuit. To see how this is done, we may set $G_c(s) = E_o(s)/E_i(s)$, and then obtain by cross multiplication,

$$E_o(s)(s^2 + 1.73\omega_0 s + 2.15\omega_0^2) = E_i(s)(s^2 + 11s + 10)\omega_0^3$$

or

$$E_o(s)\left(1 + \frac{1.73\omega_0}{s} + \frac{2.15\omega_0^2}{s^2}\right) = E_i(s)\left(1 + \frac{11}{s} + \frac{10}{s^2}\right)\omega_0^3.$$

Since ω_0 is arbitrary, we may assume $\omega_0 = 5$, which finally gives

$$E_o(s) = 125E_i(s)\left(1 + \frac{11}{s} + \frac{10}{s^2}\right) - E_o(s)\left(\frac{8.65}{s} + \frac{53.75}{s^2}\right).$$

This equation leads directly to the operational amplifier circuit. The circuit requires eight operational amplifiers, but their small size and low cost make this quite reasonable.

Example 2

We will now demonstrate that somewhat better results can be obtained by making use of the $H(s)$ in Fig. 11. We will assume $H(s) = \beta s$, so this comprises derivative feedback. We will also assume the cascade compensation function has the form

$$G_c(s) = \frac{K_c(s+1)}{s+P}$$

so one of the plant function poles is cancelled.

The plant function of Eq. (9) still applies. Then the system forward function becomes

$$\begin{aligned} G(s) &= \frac{G_c(s)G_p(s)}{1 + G_c(s)G_p(s)H(s)} \\ &= \frac{10K_c}{s[s^2 + (10+P)s + 10P + 10K_c\beta]}. \end{aligned}$$

If we now equate this $G(s)$ term by term with the $G(s)$ of Eq. (10), we obtain

$$\begin{aligned} \omega_0^3 &= 10K_c \\ 1.73\omega_0 &= 10+P \\ 2.15\omega_0^2 &= 10P + 10K_c\beta. \end{aligned} \tag{11}$$

It is now found that ω_0 must be larger than 5.78 to avoid a negative P . We will assume $\omega_0 = 10$, which results in $P = 7.3$, $K_c = 100$ and $\beta = 1.42$. The cascade compensation function which results in an optimum system is then

$$G_c(s) = \frac{100(s+1)}{s+7.3}.$$

This cascade compensation function can be readily realized by use of a standard phase lead type passive network in cascade with an amplifier with a gain of 97.8. The example illustrates the remarkable effect which can often be obtained by the addition of a simple feedback function.

It may be observed that the coefficients $A_1 = 1.73$ and $A_2 = 2.15$ in Eq. (10) needed to be realized together with the arbitrary ω_0 . There were three unknowns, K_c , P and β . It was possible to write three equations in these unknowns, as in (11), which made it possible to solve for the unknowns. In general, we must write as many independent equations as there are unknowns.

The output response of this system for a step displacement input may be read from Fig. 2, with $\alpha = 0$. Since $\omega_0 = 10$, the abscissa for this curve is $10t$. From Fig. 8 we also note that the ITAE criterion value for $\omega_0 = 1$ is $J = 3$, but in this system $\omega_0 = 10$, and we find from reference to Fig. 6 that the ITAE value for the example system is $J = 0.02$.

Example 3

Let us assume $G_p(s) = 5/(s(s+1))$ and $G_c(s) = (K_c(s+z))/(s+P)$ in Fig. 10, with $H(s) = 0$.

Then the forward function is

$$\begin{aligned} G(s) &= G_c(s)G_p(s) = \frac{5K_c(s+z)}{s(s+1)(s+P)} \\ &= \frac{5K_c(s+z)}{s[s^2 + (1+P)s + P]}. \end{aligned}$$

The system is evidently third order and we will assume Eq. (2) applies with $n = 3$. A type 1 optimum system is desired. Since $G(s)$ has a finite zero, an $F(s)$ with $\alpha \neq 0$ is required and $F(s)$ becomes

$$F(s) = \frac{C(s)}{R(s)} = \frac{\alpha A_2 \omega_0^2 s + \omega_0^3}{s^3 + A_1 \omega_0 s^2 + A_2 \omega_0^2 s + \omega_0^3}.$$

Then

$$G(s) = \frac{F(s)}{1 - F(s)} = \frac{\alpha A_2 \omega_0^2 s + \omega_0^3}{s[s^2 + A_1 \omega_0 s + A_2(1 - \alpha)\omega_0^2]}.$$

Now equating terms in the two $G(s)$ expressions,

$$\begin{aligned}5K_c z &= \omega_0^3 \\5K_c &= \alpha A_2 \omega_0^2 \\1 + P &= A_1 \omega_0 \\P &= A_2(1 - \alpha)\omega_0^2.\end{aligned}$$

We note that $P = A_1 \omega_0 - 1 = A_2(1 - \alpha)\omega_0^2$, thus if we choose α arbitrarily, A_1 and A_2 are determined and we can solve for ω_0 . Thus ω_0 is not arbitrary, but has the value 3.69, when we let $\alpha = 0.9$, and we obtain $A_1 = 3.86$ and $A_2 = 6.4$. The other quantities to be solved for, then become $P = 13.26$ per sec, $K_c = 15.69$ and $z = 0.64$ per sec. The compensation function can evidently be realized by use of a passive phase lead type network.

Although this is a type 1 system, α is quite large, thus the system characteristics will approach those of a type 2 system. The output response to a unit step input is given in Fig. 2.

Example 4

Here we consider the possibilities for designing an optimum regulator, rather than a follower system or servomechanism.

In Fig. 11 we will assume

$$G_p(s) = \frac{1}{1+s} \quad \text{and} \quad G_c(s) = \frac{K(s+z)}{s}$$

with $H(s) = 0$ and $D \neq 0$.

In a regulating system, the input R is assumed to be constant, and the principle purpose is to minimize the effect of the disturbance D .

If we assume D is an impulse function so that $D(s) = 1$, the error transform is

$$\varepsilon(s) = \frac{G_p(s)}{1 + G_c(s)G_p(s)} = \frac{s}{s^2 + (K+1)s + Kz}.$$

The objective is to determine the values of K and z required for an optimum response. In order to apply the standard forms to this problem, we first need to find a follower type system which is an equivalent system in the sense of having the same error transform as above due to an input R . To do this we equate the error transform above to $R(s)/[1 + G(s)]$ where $G(s)$ is the forward function of the unity feedback follower system and $R(s)$ is its input. We then find that if

$$G(s) = \frac{(K+1)s + Kz}{s^2} \quad \text{with} \quad R(s) = \frac{1}{s}$$

the desired result is obtained. The follower system is evidently type 2 and of second order. Its overall function is

$$F(s) = \frac{C(s)}{R(s)} = \frac{(K+1)s + Kz}{s^2 + (K+1)s + Kz}.$$

From Table II, the optimum second order type 2 system has the transfer function

$$\frac{C(s)}{R(s)} = \frac{350\omega_0 s + \omega_0^2}{s^2 + 350\omega_0 s + \omega_0^2}.$$

By equating terms, we obtain

$$K + 1 = 350\omega_0$$

$$Kz = \omega_0^2.$$

If we assume $\omega_0 = 10$, we find that $K = 1499$ and $z = 0.002858$ which are the parameter values for the compensator of the optimum regulator.

Unfortunately, standard forms suitable for all cases are not available, thus a design procedure like this cannot be applied to all regulator problems.

Example 5

In this example, we consider the design of a type 2 optimum system, using the system configuration shown in Fig. 12. We will assume the plant function to be

$$G_p(s) = \frac{300}{s(s+7)(s+36)}$$

and the cascade compensation function to be

$$G_c(s) = \frac{K_c(s+7)(s+36)}{(s+p_1)(s+p_2)}$$

thus the plant poles are cancelled and in effect replaced by the poles $s = -p_1$ and $s = -p_2$.

The feedback function $H(s)$ results from a tachometer in cascade with an operational amplifier circuit and has the form

$$H(s) = \frac{\beta s}{s + A}$$

where $A = 1/RC$ and R and C are elements in the operational amplifier feedback path.

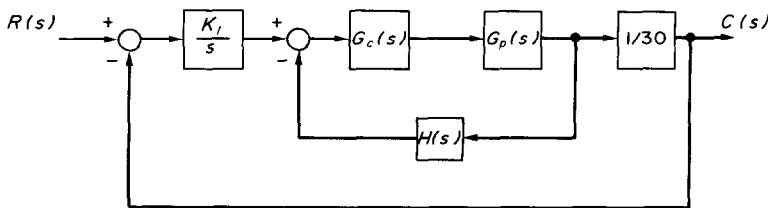


FIG. 12. Block diagram for feedback control system of Example 5.

It is expected that the system will be of fifth order, thus using the standard form for $n=5$ in Table II,

$$F(s) = \frac{C(s)}{R(s)} = \frac{1718\omega_0^4 s + \omega_0^5}{s^5 + 20.34\omega_0 s^4 + 183.1\omega_0^2 s^3 + 870\omega_0^3 s^2 + 1718\omega_0^4 s + \omega_0^5}$$

and the forward function must be

$$G(s) = \frac{F(s)}{1 - F(s)} = \frac{1718\omega_0^4(s + 5.82 \times 10^{-4}\omega_0)}{s^2(s^3 + 20.3\omega_0 s^2 + 183.1\omega_0^2 s + 870\omega_0^3)}$$

From the block diagram, we find that the forward function is

$$G(s) = \frac{10K_1K_c(s + A)}{s^2[s^3 + (p_1 + p_2 + A)s^2 + (p_1p_2 + p_1A + p_2A)s + p_1p_2A + 300K_c\beta]}$$

By comparison of the two expressions for $G(s)$, we obtain the equations

$$\begin{aligned} A &= 5.82 \times 10^{-4}\omega_0 \\ 10K_1K_c &= 1718\omega_0^4 \\ p_1 + p_2 + A &= 20.3\omega_0 \\ p_1p_2 + A(p_1 + p_2) &= 183.1\omega_0^2 \\ p_1p_2A + 300K_c\beta &= 870\omega_0^3 \end{aligned}$$

It will be noted that the choice of a value for A immediately determines ω_0 . It was found convenient to let $RC=362$, so $A=0.00276$, from which we find $\omega_0=4.74$. It was then found that $K_1K_c=8.67 \times 10^4$, $p_1p_2=4113.5$, $p_1+p_2=96.2$ and $K_c\beta=308.8$. Assuming a value of 10^4 for K_c resulted in $\beta=0.0308$ and $K_1=8.67$.

The cascade and feedback compensation functions for the system are now known. It may be observed that the cascade compensation function has complex poles, thus this function must be realized by use of an active network.

An analog simulation for this system was developed and constructed and it was found that the output response of the system to a step function input agreed very well with the response given in Fig. 4 for $\alpha=1$.

It may be observed that the $G(s)$ function has a very small zero $z = -5.82 \times 10^{-4}\omega_0$, which with $\omega_0=4.74$ becomes $z = -0.002759$. We have found that small zeros of this kind are characteristic of type 2 optimum systems. Furthermore, we have found that the $F(s)$ function in such a system has a pole of very nearly the same magnitude. In the present case this pole value is -0.0027574 .

Since the cited pole and zero values are so nearly the same, it might seem that these could be cancelled. However it is soon found that such cancellation is unacceptable, since such cancellation reduces the system to a type 1 system.

Although the small zero must be retained, it is found that the ITAE value is quite insensitive to the actual zero value. It appears that the zero value can be changed by a rather large increment, and the system will remain essentially optimum. The essential point is that the zero must be retained, but its value is

not critical. The insensitivity to the zero value is due to the fact that the small zero is always accompanied by a pole of $F(s)$ of very nearly the same value.

As previously indicated, the output response to a unit step input for this example system is given in Fig. 4 for $\alpha = 1$. Since $\omega_0 = 4.74$, the abscissa will be $4.74t$.

From Fig. 10 we note that the ITAE value for $\alpha = 1$ is evidently less than $J = 0.1$. Figure 6 shows that an increase in ω_0 from 1 to 4.74 results in a decrease in the ITAE value by the factor of $0.09/2 = 0.045$. Thus the ITAE value for the example system must be less than $J = 0.0045$.

VIII. Conclusions

A new simulation and optimization procedure has been developed, making use of the power of the modern day digital computer. By use of this procedure, it was found possible to develop standard forms for the design of optimum feedback systems. For the lower order systems, these standard forms agreed with those developed much earlier by Graham and Lathrop. For the higher order systems and in particular type 2 systems, the new standard forms were found to be quite different from those presented by the cited authors.

We call attention to the responses for $\alpha = 1$ (type 2 systems) in Figs. 1–4 and invite comparison with the responses in Ref. (1) Fig. 24. Such comparison indicates that the ITAE values for our systems are much smaller than those of Graham and Lathrop, and clearly shows that the systems of Ref. (1) cannot be optimum. The standard forms of Ref. (1) Table VI are therefore improved.

With respect to type 1 systems with $\alpha = 0$, Figs. 1–4 can be compared with Ref. (1) Fig. 22. Here we must admit that there are no great differences between the two sets of figures. When we compare the standard forms in our Table I with those in Ref. (1) Table V, we do see that there are appreciable differences for systems of fourth order and higher, but systems of second and third order agree quite well.

It must be emphasized here that although we disagree with some of the Graham and Lathrop results, we have the highest respect for these gentlemen and the work they did. Our better results are due only to the modern day availability of improved simulation and optimization techniques utilizing the digital computer.

It should be evident that the procedures for designing optimum feedback systems using our standard forms are quite simple and straightforward. Optimum feedback systems can be readily designed if a sufficient number of system parameters are available for adjustment. The designer need not always be content with merely meeting a set of specifications since it is now possible to design the best possible system of any given configuration.

The results given in this paper for type 1 systems with the α range $0 < \alpha < 1$ are entirely new and are presented here for the first time.

In this paper we have referred to the systems we have developed as optimum systems. We have done so based on our belief that they are optimum; however we must concede at this point that we know of no procedure for proving that

any particular system is optimum. In a sense, the most we can claim is that our systems are nearer to optimum than any systems heretofore developed. We suggest that our systems should be called optimum until some other investigators in the future are able to develop systems which have lower ITAE values than the ones we have presented.

References

- (1) D. Graham and R. C. Lathrop, "The synthesis of "optimum" transient response: criteria and standard forms", *AIEE Applications and Industry*, No. 9, pp. 273-288, 1953.
- (2) R. C. Lathrop and D. Graham, "The transient performance of servomechanisms with derivative and integral control", *AIEE Applications and Industry*, No. 11, pp. 10-17, 1954.
- (3) D. Graham and R. C. Lathrop, "The influence of time scale and gain on criteria for servomechanism performance", *AIEE Applications and Industry*, No. 13, pp. 153-158, 1954.
- (4) A. Nassar, "Optimum feedback systems based on the ITAE Criterion", M.S. thesis, North Carolina State University, 1976.
- (5) "CSMP III Continuous System Modeling Program Reference Manual", IBM, White Plains, N.Y. SH19-7001-2, 1972.
- (6) M. J. D. Powell, "An efficient method for finding the minimum of a function of several variables without calculating derivatives", *Computer J.*, Vol. 7, pp. 155-162, 1964.
- (7) H. H. Rosenbrock, "An automatic method for finding the greatest or least value of a function", *Computer J.*, Vol. 3, pp. 175-184, 1960.
- (8) R. L. Stahr, Ph.D. dissertation, Department of Chemical Engineering, University of Texas, Austin, Texas, 1973.